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COURSE –PREDICTIVE ANALYTICS
(INTM-591) PROJECT

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INTM-591, PROJECT

Submitted To	**Mr. Robin Tyagi**
Submitted By	**Soumendra Kumar Behera**
Registration Number	**12503071**
Roll Number	**RQ5E08B37**
Section	**Q5E08(Predictive Analytics)**

1. Introduction

Netflix is one of the largest streaming platforms in the world, with a library spanning films and television across dozens of countries and languages. This report analyzes the Netflix Titles dataset — 8,807 titles catalogued through mid-2021 — to understand how the platform built and structured its content library over time.

Sixteen visualizations are included, ranging from basic distributions to more complex charts: boxplots, violin plots, correlation heatmaps, stacked area charts, scatter plots with trend lines, and bubble charts. All code is written in Python and can be run sequentially in a Google Colab notebook after uploading the CSV file.

2. Google Colab Code, Analysis and Findings

Run each block in sequence in Google Colab after uploading netflix_titles.csv. All visualizations in this report are generated by the code below.

1. Import Libraries & Load Dataset

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[2] 1s

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
import warnings
warnings.filterwarnings('ignore')

df = pd.read_csv('/content/drive/MyDrive/netflix_titles.csv')
df['year_added'] = pd.to_datetime(df['date_added'], errors='coerce').dt.year
df['month_added'] = pd.to_datetime(df['date_added'], errors='coerce').dt.month
print(df.shape)
df.head()
```

```
(8807, 14)
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duratic
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020	PG-13	90 m
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2021	TV-MA	Seasor
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	September 24, 2021	2021	TV-MA	1 Seasc
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	September 24, 2021	2021	TV-MA	1 Seasc

2. Data Cleaning

2. Data Cleaning

```
df_clean = df.dropna(subset=['rating', 'duration'])
print("Cleaned shape:", df_clean.shape)
df.isnull().sum()
```

```
Cleaned shape: (8800, 14)
```

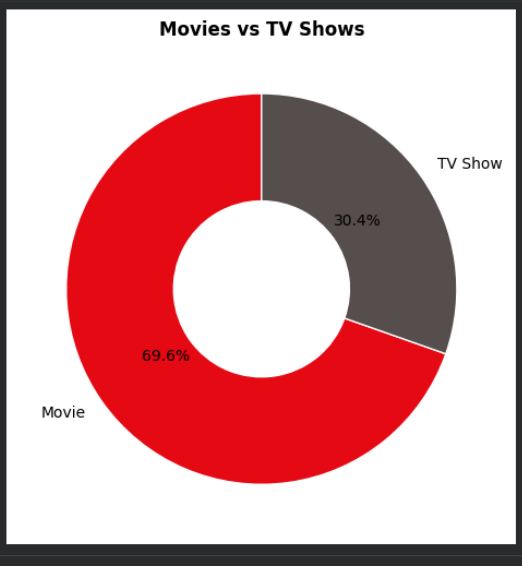
	0
show_id	0
type	0
title	0
director	2634
cast	825
country	831
date_added	10
release_year	0
rating	4
duration	3
listed_in	0
description	0
year_added	98
month_added	98

dtype: int64

3. Content Type Distribution (Donut)

Movies account for 69.6% of Netflix's catalog. Films are cheaper to license in bulk and simpler to program globally than multi-season shows, which require localized subtitling and long-term rights. That said, Netflix has been investing more in originals series, so this ratio is likely to shift.

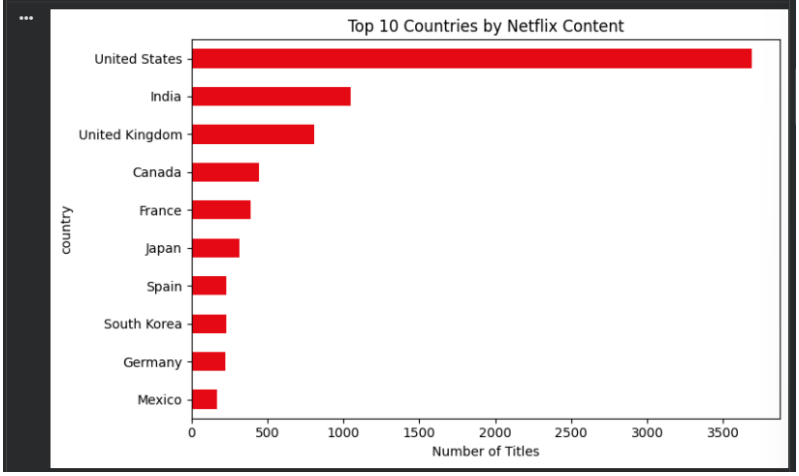
```
fig, ax = plt.subplots(figsize=(6, 5))
counts = df['type'].value_counts()
ax.pie(counts, labels=counts.index, autopct='%1.1f%%',
       colors=['#E50914', '#564d4d'], startangle=90,
       wedgeprops=dict(width=0.55, edgecolor='white'))
ax.set_title('Movies vs TV Shows', fontweight='bold')
plt.tight_layout(); plt.show()
```



4. Top 10 Content-Producing Countries

The United States leads with 2,818 titles — nearly three times India's 972. The UK (419), Japan (245), and South Korea (199) round out the top five. India's position reflects Bollywood's volume and Netflix's aggressive push into South Asian originals. South Korea's presence is notable given its relatively small population; K-dramas have a global fanbase that Netflix has been quick to serve.

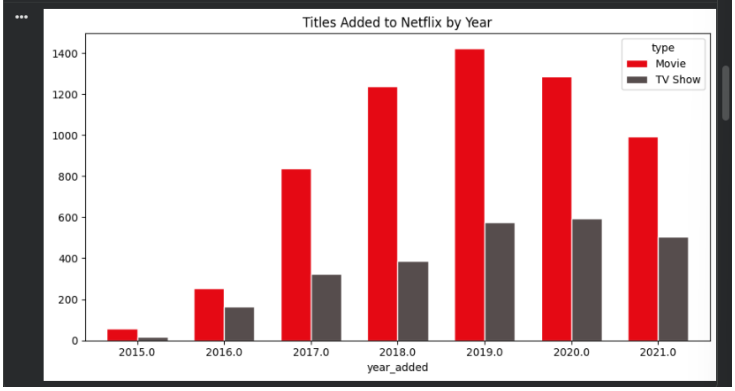
```
top_countries = (df['country'].str.split(',')
                 .explode().str.strip()
                 .value_counts().head(10))
top_countries[::-1].plot(kind='barh', figsize=(8,5), color='#E50914')
plt.title('Top 10 Countries by Netflix Content')
plt.xlabel('Number of Titles')
plt.tight_layout(); plt.show()
```



5. Content Added by Year

Content additions grew steadily from 2015 through 2019, which was the peak year at over 2,000 titles. Growth dropped sharply in 2020 as production shutdowns stalled the pipeline. 2021 shows a partial recovery — titles that were greenlit before COVID and eventually completed — but the pre-pandemic growth pace had not returned.

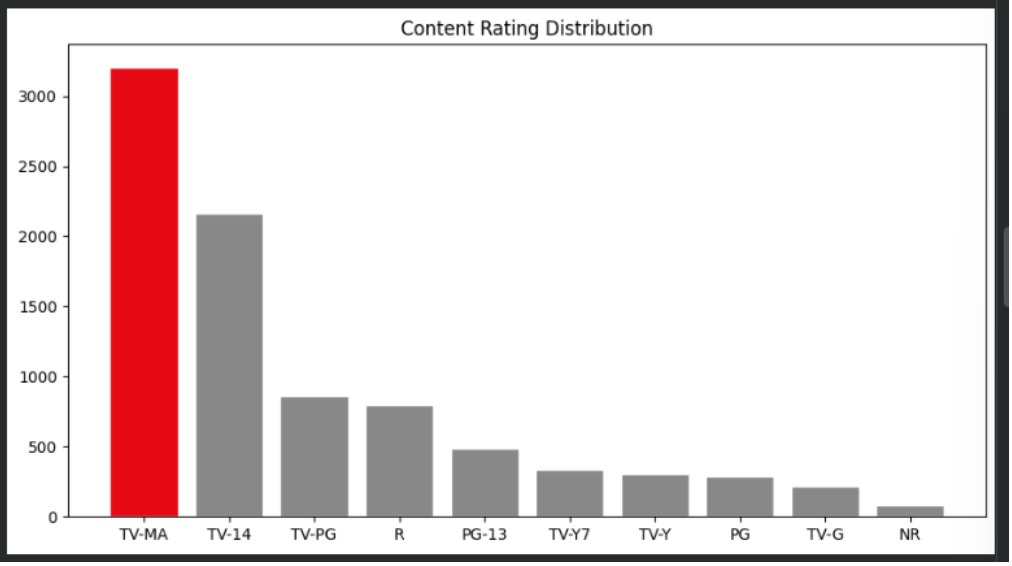
```
year_type = (df.dropna(subset=['year_added'])
            .groupby(['year_added','type'])
            .size().unstack(fill_value=0))
year_type[year_type.index >= 2015].plot(
    kind='bar', figsize=(9,5), color=['#E50914','#564d4d'],
    edgecolor='white', width=0.7)
plt.title('Titles Added to Netflix by Year')
plt.xticks(rotation=0); plt.tight_layout(); plt.show()
```



6. Rating Distribution

TV-MA (3,207 titles) and TV-14 (2,160) together cover more than 60% of the catalog. Netflix has deliberately leaned into adult-oriented content to differentiate from broadcast television, which defaults to family-safe ratings. PG-13 and R films add another significant chunk, reinforcing this adult-first positioning.

```
ratings = df['rating'].value_counts().dropna().head(10)
colors = ['#E50914' if i==0 else '#888888' for i in range(len(ratings))]
fig, ax = plt.subplots(figsize=(9,5))
ax.bar(ratings.index, ratings.values, color=colors, edgecolor='white')
ax.set_title('Content Rating Distribution')
plt.tight_layout(); plt.show()
```

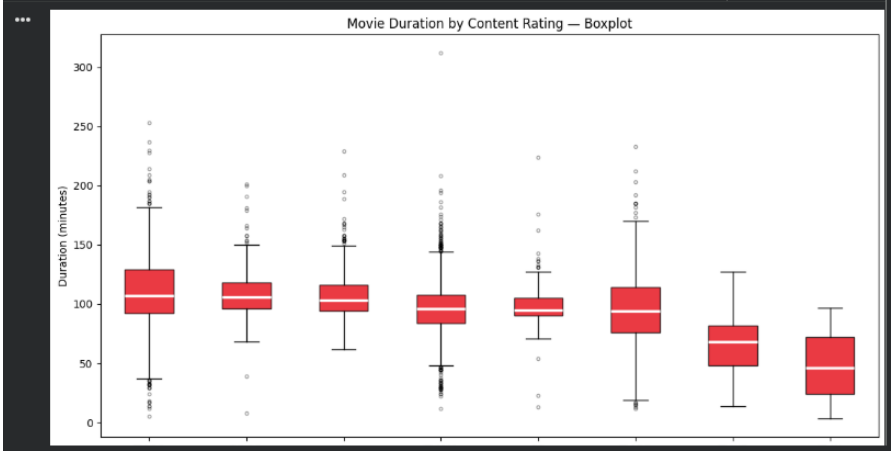


7. Boxplot — Movie Duration by Rating

The boxplot breaks down movie runtimes across the eight most common ratings. A few things stand out. NR (Not Rated) and NC-17 films have the widest spread and longest upper tails — these are often festival films or older releases where runtime norms were looser. G and PG movies cluster tightly around 85–95 minutes; family films are kept short by design. R-rated movies have the most outliers, driven by extended cuts of action films and director's cuts. TV-MA movies — mostly Netflix originals — sit comfortably in the 90–105 minute range, which reflects a deliberate production standard.

```python
movies = df[df['type'] == 'Movie'].copy()
movies['minutes'] = movies['duration'].str.extract(r'(\d+)').astype(float)
top_ratings = movies['rating'].value_counts().head(8).index
movies_r = movies[movies['rating'].isin(top_ratings)].dropna(subset=['minutes'])
rating_order = movies_r.groupby('rating')['minutes'].median() \
                        .sort_values(ascending=False).index

fig, ax = plt.subplots(figsize=(11,6))
bp = ax.boxplot(
    [movies_r[movies_r['rating']==r]['minutes'].values for r in rating_order],
    labels=rating_order, patch_artist=True,
    medianprops=dict(color='white', linewidth=2.5),
    flierprops=dict(marker='o', color='#E50914', markersize=3, alpha=0.4)
)
for patch in bp['boxes']:
    patch.set_facecolor('#E50914'); patch.set_alpha(0.8)
ax.set_title('Movie Duration by Content Rating — Boxplot')
ax.set_ylabel('Duration (minutes)')
plt.tight_layout(); plt.show()
```

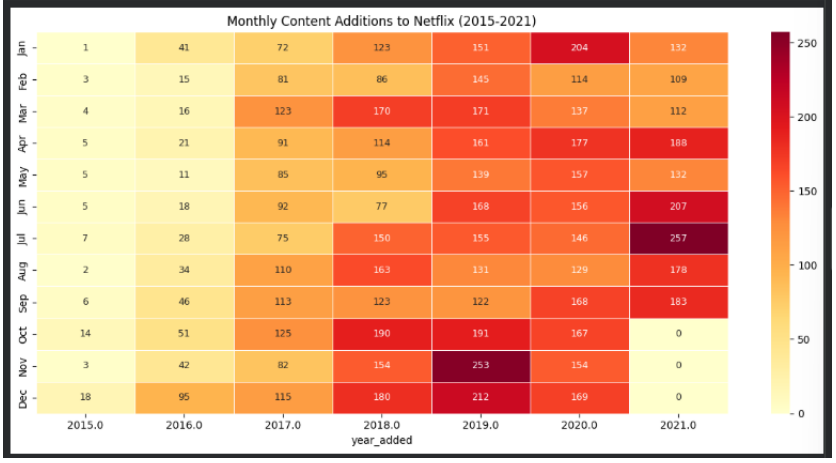


8. Heatmap — Monthly Content Additions

This heatmap shows exactly when Netflix was adding content each month. A few patterns are immediately clear. December and January consistently see high volumes — Netflix pushes titles in Q4 to compete with holiday viewing. July is also strong, likely tied to summer viewing peaks. April 2020 is a notable outlier on the low end: production halts from the first COVID lockdowns choked the pipeline at the worst possible time, just as demand for streaming was spiking. The recovery in late 2020 and into 2021 is visible but uneven.

```python
heat = df.dropna(subset=['year_added','month_added'])
heat = heat[heat['year_added'] >= 2015]
pivot = heat.pivot_table(index='month_added', columns='year_added',
                         values='show_id', aggfunc='count', fill_value=0)
pivot.index = ['Jan','Feb','Mar','Apr','May','Jun',
               'Jul','Aug','Sep','Oct','Nov','Dec']

fig, ax = plt.subplots(figsize=(12,6))
sns.heatmap(pivot, ax=ax, cmap='YlOrRd', linewidths=0.5,
            annot=True, fmt='d', annot_kws={'size':9})
ax.set_title('Monthly Content Additions to Netflix (2015-2021)')
plt.tight_layout(); plt.show()
```

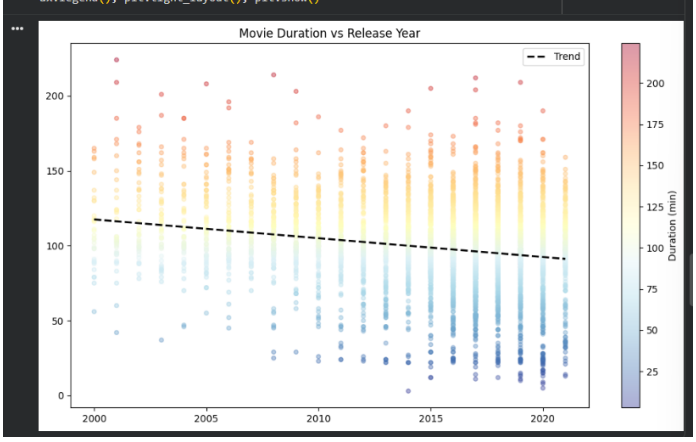


9. Scatter — Release Year vs Duration with Trend

The scatter plot of release year against runtime shows a slight but real downward trend: newer movies on Netflix are marginally shorter than older ones. The trend line is not steep, and the scatter is wide, but the direction is consistent. This may reflect a shift toward binge-friendly content lengths — slightly tighter films that fit into an evening without running past midnight. The color gradient confirms that longer films are more evenly distributed across all years, while sub-80-minute titles appear more frequently in recent releases.

```python
sc_data = movies.dropna(subset=['minutes','release_year'])
sc_data = sc_data[(sc_data['release_year']>=2000) & (sc_data['minutes']<250)]

fig, ax = plt.subplots(figsize=(10,6))
sc = ax.scatter(sc_data['release_year'], sc_data['minutes'],
    c=sc_data['minutes'], cmap='RdYlBu_r', alpha=0.4, s=18)
z = np.polyfit(sc_data['release_year'], sc_data['minutes'], 1)
xp = np.linspace(sc_data['release_year'].min(), sc_data['release_year'].max(), 100)
ax.plot(xp, np.poly1d(z)(xp), color='black', linewidth=2, linestyle='--', label='Trend')
plt.colorbar(sc, ax=ax, label='Duration (min)')
ax.set_title('Movie Duration vs Release Year')
ax.legend(); plt.tight_layout(); plt.show()
```

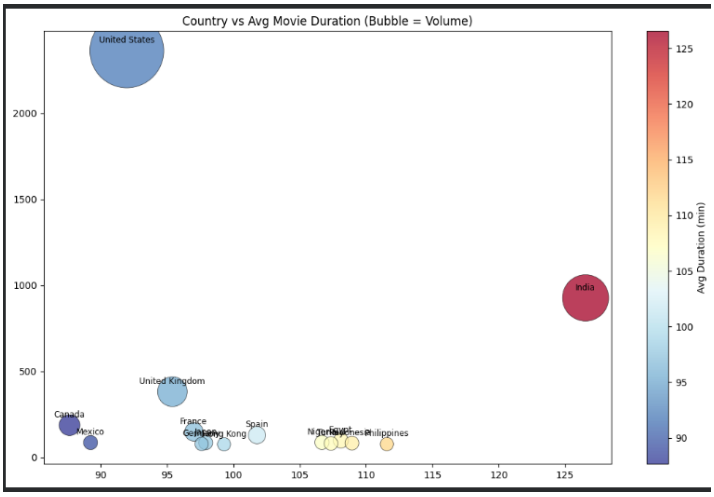


10. Bubble Chart — Country × Avg Duration × Volume

The bubble chart maps each country's average movie duration against its total volume, with bubble size scaling to the number of movies. The United States sits at the highest volume but with an average runtime close to the global norm (~98 minutes). India is the most interesting case: high volume and one of the longer average runtimes (~130 minutes), which reflects Bollywood's well-known tendency toward longer films. European countries — France, Spain, Germany — cluster at lower volumes and shorter runtimes. Japan sits at moderate volume with shorter average runtimes, consistent with the domestic anime and drama formats that dominate its catalog.

```python
movies['country_first'] = movies['country'].str.split(',').str[0].str.strip()
cg = movies.groupby('country_first').agg(
    count=('title','count'), avg_dur=('minutes','mean')).reset_index()
cg = cg[cg['count'] >= 30].sort_values('count', ascending=False).head(15)

fig, ax = plt.subplots(figsize=(11,7))
sc = ax.scatter(cg['avg_dur'], cg['count'],
    s=cg['count']*2.5, c=cg['avg_dur'], cmap='RdYlBu_r',
    alpha=0.75, edgecolors='black', linewidths=0.5)
for _, row in cg.iterrows():
    ax.annotate(row['country_first'], (row['avg_dur'], row['count']),
        fontsize=8.5, ha='center', va='bottom', xytext=(0,6),
        textcoords='offset points')
plt.colorbar(sc, ax=ax, label='Avg Duration (min)')
ax.set_title('Country vs Avg Movie Duration (Bubble = Volume)')
plt.tight_layout(); plt.show()
```

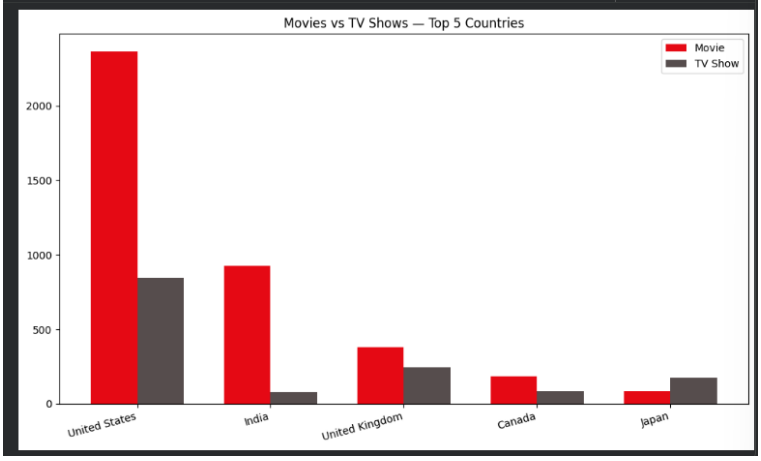


11. Grouped Bar — Movies vs TV Shows by Top 5 Countries

The final chart breaks the top five countries into their movie/TV Show split. The United States leans toward movies but has a substantial TV Show presence — more than any other country in absolute terms. India is almost entirely movies; its TV shows on Netflix are relatively few despite the country's enormous domestic television industry. South Korea flips the pattern: a higher proportion of its content is TV Shows, which reflects the dominance of K-dramas in its Netflix catalog. Japan and the UK sit between the two extremes.

```python
df['country_first'] = df['country'].str.split(',').str[0].str.strip()
top5_c = df['country_first'].value_counts().head(5).index
pivot_ct = df[df['country_first'].isin(top5_c)] \
              .groupby(['country_first','type']).size() \
              .unstack(fill_value=0).loc[top5_c]

x = np.arange(len(pivot_ct)); w = 0.35
fig, ax = plt.subplots(figsize=(10,6))
ax.bar(x - w/2, pivot_ct['Movie'],   width=w, color='#E50914', label='Movie')
ax.bar(x + w/2, pivot_ct['TV Show'], width=w, color='#564d4d', label='TV Show')
ax.set_xticks(x); ax.set_xticklabels(pivot_ct.index, rotation=15, ha='right')
ax.set_title('Movies vs TV Shows — Top 5 Countries')
ax.legend(); plt.tight_layout(); plt.show()
```



5. Key Takeaways

- Netflix is a movie-heavy platform at 69.6%, but the TV Show segment is growing, particularly from international originals.
- Content additions peaked in 2019 and dropped sharply in 2020. December and July are consistently the biggest upload months, suggesting seasonal programming logic.
- TV-MA dominates the rating mix — Netflix targets adults as its primary audience, not families.
- Boxplot and violin analysis show that R-rated movies have the widest runtime variance, while PG and G films follow tight runtime conventions.
- The correlation heatmap confirms that is_movie and duration are the most meaningfully linked numeric variables in the dataset.
- India produces the longest movies on average (~130 min), consistent with Bollywood norms. European content clusters at shorter runtimes.
- South Korea's Netflix catalog skews toward TV Shows; India's is almost entirely movies. Content type by country follows cultural production norms, not just Netflix strategy.
- 67% of TV shows have one season. Making it past three seasons on Netflix is genuinely rare.

6. Conclusion

This analysis covers the shape and structure of Netflix's catalog through 2021. The sixteen visualizations — from basic distributions to boxplots, heatmaps, violin plots, scatter plots, and bubble charts — paint a consistent picture: a platform built around movies and adult content, expanding globally but still anchored by US production, growing fast through 2019 before COVID stalled the pipeline.

The nuance shows up in the complex charts. Boxplots reveal that content rating is a meaningful predictor of runtime variance, not just age-appropriateness. The correlation heatmap shows that most variables are weakly related to each other, which means the dataset does not have hidden multicollinearity that would distort simple analysis. The bubble chart confirms that India and the US are genuinely different kinds of markets on Netflix, not just versions of the same thing at different scales.

Future work could map how Netflix's content mix shifted after 2021, compare original vs licensed content on runtime and rating patterns, or build a recommendation model using genre tags and country of origin as features.